REPORT ON STUDENT WORK EXPERIENCE PROGRAM (SWEP) II

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DEPARTMENT: COMPUTER ENGINEERING

ORGANIZATION: UNIVELCITY

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AFE BABALOLA UNIVERSITY, ADO-EKITI

EKITI STATE

DEDICATION

I dedicate this report to Almighty God, whose guidance and wisdom have been a steadfast source of support throughout the duration of this program.

To my esteemed family, whose unwavering encouragement and support have been instrumental in my academic and professional pursuits.

To my distinguished lecturers and supervisors, whose invaluable insights and mentorship have significantly contributed to my growth and development.

Finally, to my colleagues and friends, whose companionship and collaboration have enriched this experience.

CERTIFICATION

This is to certify that OYEWUMI VICTOR OLUMIDE of the computer engineering department, with Matriculation Number 22/ENG02/082, has successfully completed the Student Work Experience Programme (SWEP) II at UNIVELCITY. This program was undertaken as part of the requirements for the award of a degree from Afe Babalola University Ado-Ekiti.

This report has been duly reviewed and approved by the undersigned, and it reflects the activities and knowledge gained during the SWEP.

_____ _____

Dr. Pascal Chimeremeze Chiadighikaobi Date

SWEP SUPERVISOR

_____ _____

Engr. Dr. Ogunlade Date

SIWES Coordinator

Computer & Electrical/Electronic Engineering

_____ _____

Engr. Dr. Ojo Date

Head of Department

Computer & Electrical/Electronic Engineering

ACKNOWLEDGEMENT

I wish to express my profound gratitude to Almighty God for His continual guidance, strength and sustenance throughout my SWEP in Univelcity.

My profound appreciation goes to the Provost of the College of Engineering Engr. Prof Femi Onibonoje and the Head of Department, Computer and Electrical/Electronic Engineering Engr. Dr. Ojo

I am deeply indebted to my academic supervisors, Dr. Pascal Chimeremeze Chiadighikaobi, whose expertise, constructive feedback, and unwavering support were crucial to the successful completion of this report.

My sincere appreciation goes to the management and staff of UNIVELCITY for providing me with the opportunity to undertake my Student Work Experience Program (SWEP) within their esteemed organization. Their mentorship, resources, and practical insights have significantly enhanced my understanding of the industry.

I also wish to acknowledge the support and encouragement of my family, whose sacrifices and belief in my abilities have been a constant source of motivation.

Finally, I extend my thanks to my colleagues and friends, whose collaboration, advice, and shared experiences made this journey both enriching and enjoyable.

ABSTRACT

The Student Work Experience Programme (SWEP) is an industrial training initiative designed to bridge the gap between classroom theory and practical application. This report documents my SWEP II experience at Univelcity, where I received intensive training in modern software engineering practices. The program was structured in two phases: frontend development and backend development.

The frontend phase covered the fundamentals of HTML, CSS, JavaScript, React, and Tailwind CSS. During this period, I developed several practical projects including a task manager, BMI calculator, countdown timer, and a fully functional portfolio website. The backend phase introduced server-side technologies such as Node.js, Express, MongoDB, and Mongoose, culminating in the development of a Subscription Tracker Application with user authentication, and data persistence. Tools such as Git/GitHub, Postman, Netlify, and Vercel were also employed for version control, API testing, and project deployment.

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CHAPTER ONE
INTRODUCTION

1.1 INTRODUCTION TO STUDENT WORK EXPERIENCE PROGRAM (SWEP)

The Student Work Experience Programme (SWEP) was introduced in the late 20th century, around the 1980s, as part of the curriculum in various technical and engineering institutions. The program was developed in response to the increasing need for practical experience in educational settings, particularly in fields like engineering, where hands-on skills are crucial.

The inception of SWEP aimed to address the gap between theoretical knowledge and practical application, which was becoming more apparent as industries advanced. By the early 1990s, SWEP had been adopted by numerous institutions globally, integrating it as a mandatory component of engineering and technical education.

Over the years, SWEP has evolved to include a wide range of practical experiences, from on-campus projects to collaborations with industry partners. These experiences provide students with the opportunity to apply classroom knowledge to real-world situations, develop critical problem-solving skills, and gain exposure to professional environments.

The program continues to be an essential part of engineering and technical education today, ensuring that graduates are not only academically proficient but also industry-ready. As technology and industry demands continue to change, SWEP remains a vital tool for preparing students to meet these challenges effectively.

1.2 AIM OF SWEP

The aim of the Student Work Experience Program (SWEP) is to provide students with practical, hands-on experience in their field of study, bridging the gap between theoretical knowledge and real-world application. Through this program, students are

expected to develop technical and professional skills, enhance their understanding of industry practices, and gain insights into the workplace environment.

1.3 OBJECTIVES OF SWEP

1. **To Provide Practical Experience:** The primary aim of SWEP is to equip students with practical skills by engaging them in real-world projects that align with their academic studies.
2. **To Bridge the Gap Between Theory and Practice:** SWEP seeks to connect the theoretical knowledge gained in the classroom with its practical application in the field, enhancing the students' overall understanding of their discipline.
3. **To Foster Industry Readiness:** By participating in SWEP, students become more prepared for the challenges of the industry, gaining experience that makes them more competitive in the job market.
4. **To Develop Problem-Solving Skills:** The program encourages students to confront and solve practical problems, thereby honing their analytical and critical thinking abilities.
5. **To Promote Teamwork and Collaboration:** SWEP often involves group projects, which help students develop the ability to work effectively in teams, a crucial skill in any professional environment.
6. **To Enhance Professionalism and Work Ethics:** The experience gained through SWEP helps students understand the importance of professionalism, time management, and work ethics, which are essential for career success.

1.4 LOGBOOK

The logbook given to me before the commencement of my SWEP program is used to record my activities in my workplace during the period of attachment and it was checked,signed and commented on by the industry based supervisor and institution based supervisor. The logbook served multiple purposes: it provided evidence of

continuous learning, facilitated supervisor review, and acted as a reference for debugging and future development.

1.5 BRIEF HISTORY OF THE ORGANIZATION - UNIVELCITY

1.5.1 Overview

Fig 1.1: The Organization LOGO

Univelcity is a Lagos-based technology training institute dedicated to grooming world-class software engineers. Founded in 2016 by Joseph Agunbiade to address the gap between academic training and industry needs, Univelcity provides immersive, hands-on training programs that prepare students for careers in technology. Its programs are designed to ensure that students gain not only theoretical knowledge but also the practical skills required to excel in the workplace.

The institution focuses heavily on project-based learning, ensuring that every concept taught is reinforced through real-world application. Students are encouraged to build projects individually and in groups, giving them the opportunity to collaborate as they would in professional software engineering teams.

1.5.2 Vision & Mission

- **Vision:** To cultivate a community of skilled developers capable of solving real-world problems using modern software technologies.
- **Mission:** To provide practical, hands-on training that equips learners with the technical and professional skills required by the software industry.

1.5.3 Core Values

1. **Excellence**: Delivering high-quality instruction and mentorship.
2. **Practicality**: Prioritizing hands-on project work over passive learning.
3. **Collaboration**: Encouraging teamwork and peer learning.
4. **Innovation**: Staying current with technology trends and best practices.
5. **Integrity:** Maintaining professional conduct and ethical standards.

1.5.4 Nature of Services

Univelcity provides:

1. Training bootcamps in frontend, backend, and full-stack development.
2. Mentorship and portfolio coaching.
3. Project incubation and support for deploying student projects.
4. Corporate upskilling and short course offerings.

1.5.5 Tools & Platforms Commonly Used

Univelcity's practical sessions and projects typically use the following:

1. **Editors/IDEs**: Visual Studio Code, GoLand (when applicable)
2. **Version Control & Hosting**: Git, GitHub, Netlify, Vercel

3. **Backend & DB**: Node.js, Express, MongoDB Atlas, Mongoose
4. **Testing & API Tools**: Postman, HTTPie
5. **UI**: React, Tailwind CSS, CSS Modules, Heroicons
6. **Design & Prototyping**: Figma, BennettFeely (clip-paths experimenting)
7. **CI/Dev Tool**s: ESLint, Nodemon, dotenv

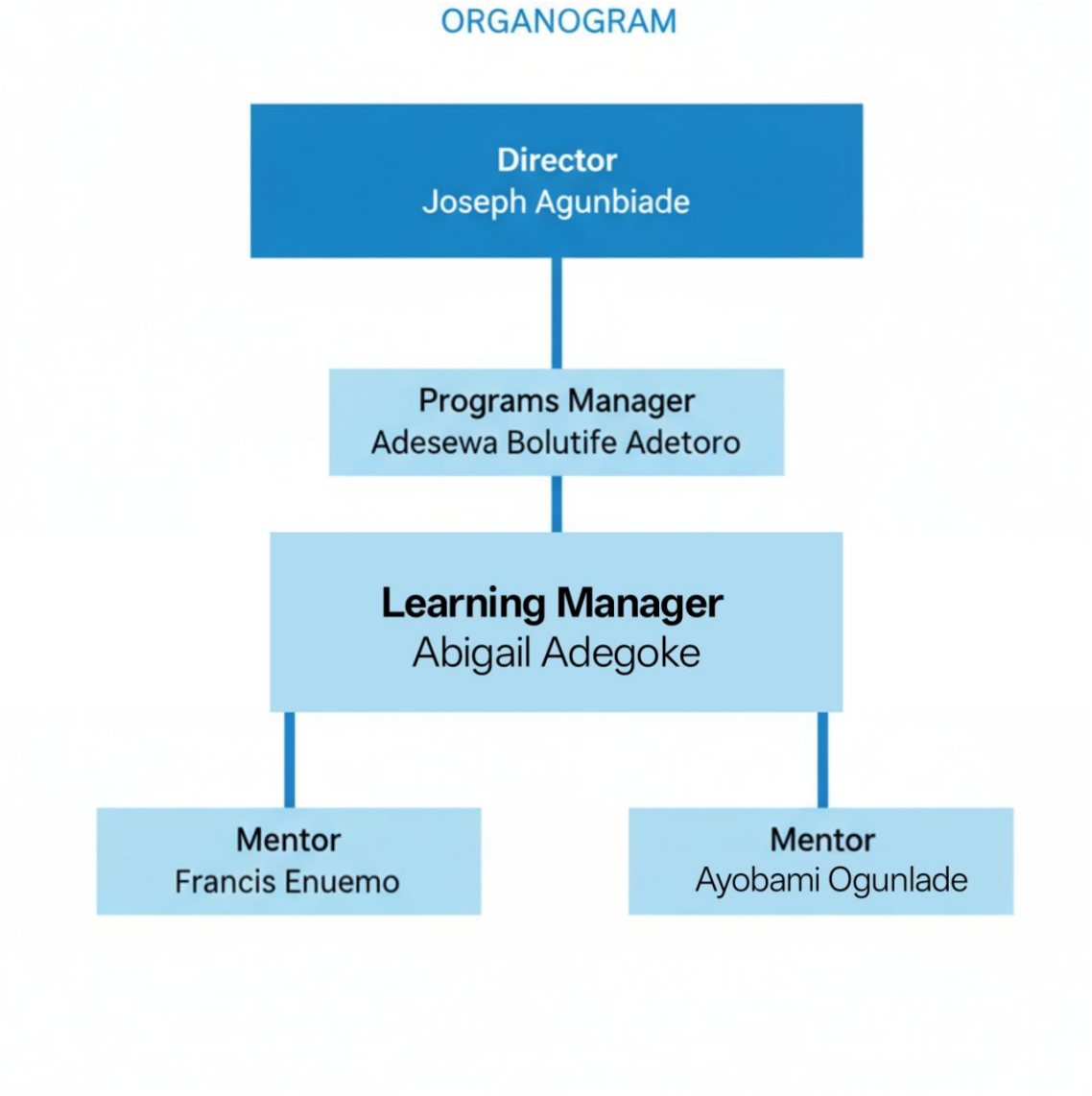
1.5.6 Products and Services

1. Software Engineering Training: Covering frontend, backend, and full-stack development.
2. Data Science and Artificial Intelligence Training.
3. Cyber Security.
4. Corporate Training Programs for organizations seeking to upskill staff.
5. Mentorship and Project Supervision for students building real-world solutions.

1.5.5 Number of Employees: The company employs a dedicated team of professionals with diverse expertise, including software engineers and instructors, product designers, learning manger, program manager, and consultants. This talented team drives the company's success and contributes to its ongoing growth and development.

1.6 ORGANIZATION STRUCTURE & PERSONNEL

Univelcity is structured around instructional leads, mentors, and administrative staff. Instructors are practicing software engineers responsible for delivering lessons and supervising projects. Mentors provide one-on-one support during assignment periods, while administrative staff handle scheduling and platform access.



1.7 SUMMARY OF INTERN'S ROLE, RESPONSIBILITIES & ACTIVITIES

As an intern at Univelcity, my roles and responsibilities included:

Core Responsibilities

- Attend all scheduled training sessions and actively participate in discussions and coding activities.
- Complete daily assignments and submit solutions via GitHub and Google Classroom.
- Collaborate with peers on group projects.
- Maintain a daily logbook documenting topics learned, tools used, and outcomes.

- Deploy completed frontend projects to hosting platforms and submit links for assessment.
- Assist in building backend APIs, setting up database clusters, and implementing routes and controllers.
- Present projects during demo sessions and incorporate feedback.

Typical Daily Activities

- Participated in instructor-led lessons and followed along with live coding demonstrations.
- Implemented UI components in React, styled them using Tailwind or CSS Modules, and tested responsiveness across device breakpoints.
- Wrote Express routes and controllers for backend functionality, tested endpoints with Postman, and integrated Mongoose models.
- Used Git for version control: initialized repositories, created feature branches, committed changes, and raised pull requests where necessary.
- Researched documentation and community resources when encountering issues, applying fixes and optimizations under mentorship.

Key Projects & Deliverables (selected)

- **Portfolio Website:** built with React and Tailwind; hosted and presented as a capstone.
- **Task Manager & Live Clock**: small DOM-based apps to practice JavaScript and event handling.
- **BMI Calculator & Countdown Timer**: DOM-interactive assignments with validation and UI feedback.
- **Commercial shopping app (slug-based)**: dynamic routing in React for product detail views with similar-product suggestions.
- **Subscription Tracker (Backend)**: Express/Mongoose application with Auth, User, and Subscription routes; JWT authentication; deployed database on MongoDB Atlas.

1.8 Expected Learning Outcomes / Competencies Gained

By the end of the attachment period, we were expected to:

- Demonstrate the ability to build responsive, accessible frontend interfaces with React and Tailwind.
- Implement RESTful API endpoints and integrate them with frontend forms and components.
- Use MongoDB/Mongoose for data persistence and perform CRUD operations via Express routes.
- Apply version control best practices using Git and GitHub.
- Deploy frontend applications to hosting platforms and document the deployment steps.
- Present technical work clearly and incorporate feedback from peers and mentors.

Overall, my role was to learn actively, apply knowledge, and demonstrate growth through consistent participation, project delivery, and documentation.

CHAPTER TWO

TRAINING PROGRAM

2.0 Overview

The training program at Univelcity was a progressive, hands-on curriculum that began with frontend fundamentals and advanced into full-stack development. The program combined instructor-led lessons, individual assignments, group projects, and capstone work. Practical tasks emphasized building real web applications, testing APIs, and deploying projects to public hosts. The training was intentionally iterative: each module (HTML → CSS → JavaScript → React → Backend/Express → MongoDB) reinforced previous lessons while introducing new tooling and best practices.

2.1 FRONTEND DEVELOPMENT

2.1.1 HTML Foundations

Topics covered: HTML document structure (<!doctype html>, <head>, <body>), semantic tags (<header>, <nav>, <main>, <section>, <article>, <footer>), block vs inline elements, lists (<ul>, <ol>, <dl>), grouping elements (, , <article>, <section>), multimedia tags (<img>, <audio>, <video>), form elements (<form>, <input>, <select>, <textarea>,<option>), and accessibility basics (alt text, labels).

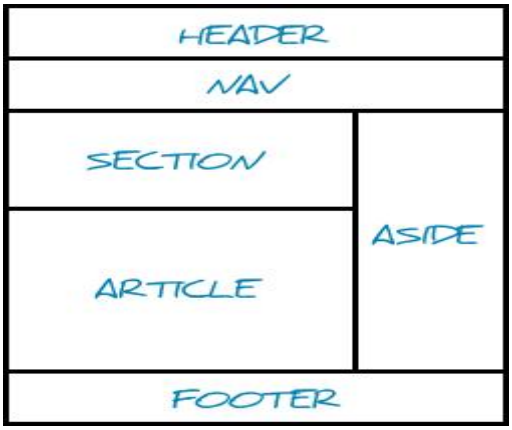


FIG 2.1: HTML page wireframe

2.1.2 CSS Basics and Best Practices

Topics covered: CSS selectors (element, class, id, descendant, child, adjacent, sibling), cascade and specificity, box model (content, padding, border, margin), units (px, em, rem, %, vw, vh), color systems (RGB, HSL, HWB), CSS preprocessors overview (LESS, SASS), Google Fonts integration, and responsive design via media queries (media screen, media print).

Practical tasks: built multiple layouts; created flags using color models to practice color theory; integrated custom fonts from Google Fonts.

Tip: Use relative units (rem) for scalable typography and min()/max() for modern responsive sizing.

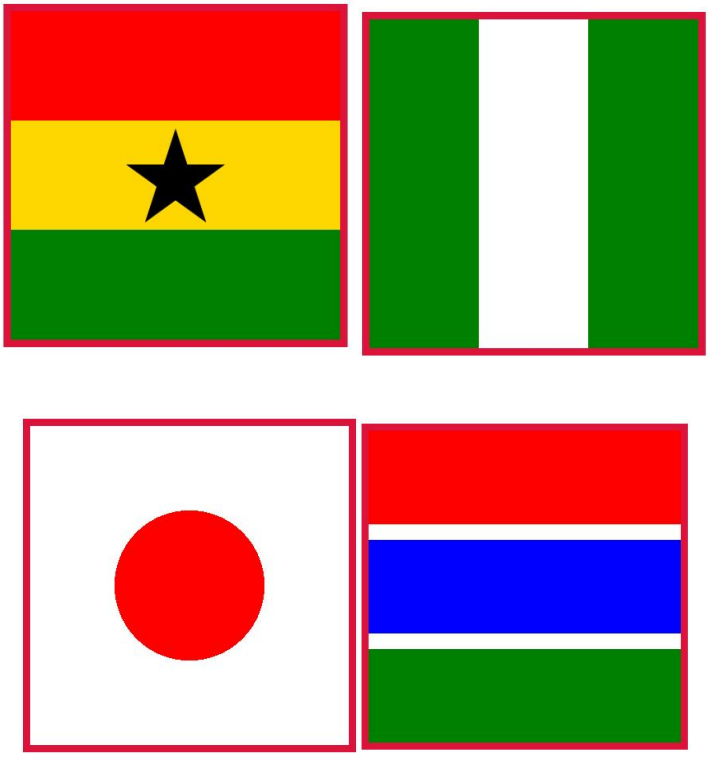


FIG 2.2: CSS flags

2.1.3 Layout Systems: Flexbox & Grid

Topics covered: CSS Grid (grid-template-columns, grid-template-rows, grid-area) and Flexbox (display:flex, justify-content, align-items), grid placement, implicit vs explicit grid tracks, and responsive patterns.

Practical tasks: designed multi-column product layouts and responsive landing sections using Grid and Flexbox.

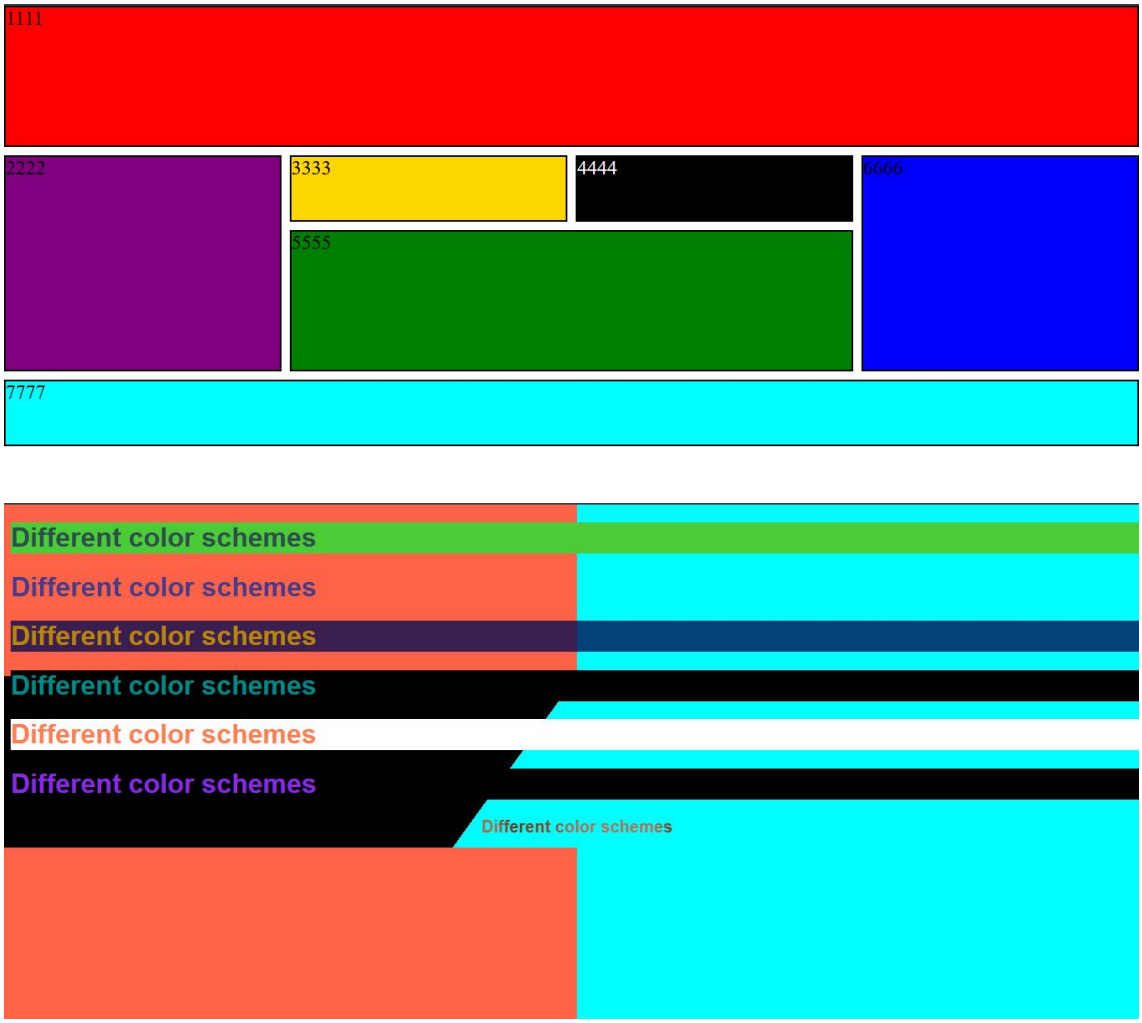


FIG 2.3: grid box image and color scheme

2.1.4 Utility-First: Tailwind CSS

Topics covered: Installing Tailwind (typical steps: “npm init” initialize node package model, npm install tailwindcss @cli), utility classes, @layer base/components/utilities, customizing theme (font weights, spacing), and creating reusable components via @layer components.

Practical tasks: converted some CSS layouts to Tailwind, created custom components using @layer components, and defined grid rows/columns using Tailwind classes.

Benefit: faster prototyping and consistent, constrained styles across the app.

2.1.5 JavaScript Fundamentals

Core topics: JS history and environment, variables (let, const, var), data types (primitive/non-primitive), string methods (length, indexOf, slice, toUpperCase, toLowerCase, split, concat, startsWith, endsWith, charAt, replace, trim, padStart, match), operators, truthy/falsy, Math functions(rand(), floor(), …), control flow (if/else statement, switch statement), loops (for, while, do...while).

Functions: declaration of function, expression, arrow functions, anonymous functions, constructor functions. Implemented examples and exercises.

Arrays: creation of array, typed examples (number arrays, string arrays), common methods in an array (push, pop, shift, unshift, splice, slice, map, filter, reduce).

DOM(Document Object Model) & Events: querySelector, querySelectorAll, getElementById, getElementsByClassName, getElementsByTagName, innerText vs innerHTML, try{} catch{}, eval().

Practical tasks & assignments:

- Task Manager (add, remove, persist to local state)
- Live clock (setInterval, DOM updates)

- BMI calculator (validate input, compute BMI, display comment)
- Countdown timer (timers, DOM updates)

FIG 2.4: Task Manager UI & BMI result screenshot

2.1.6 Introduction React

React basics: Why React, JSX, components, props, state, useState, useParams, useEffect, component-driven architecture.

Project tasks:

- Created a Next app (via npx create-next-app@latest) along with Eslint, Turbo pack and Tailwind css and renamed default files.
- Built a shopping website, added a product slug page (dyanmic route) showing full product details and similar products component that picks three random items from an array.
- Implemented navigation responsive fixes: hamburger menu dropdown(React component) for mobile, and for desktop.
- Cart functionality: added "Add to cart" button; implemented state management patterns (context / useReducer) to manage cart state.
- Replicate a portfolio website template by RIVERA
- Hosting & Deployment: pushed code to GitHub; deployed to Netlify and Vercel

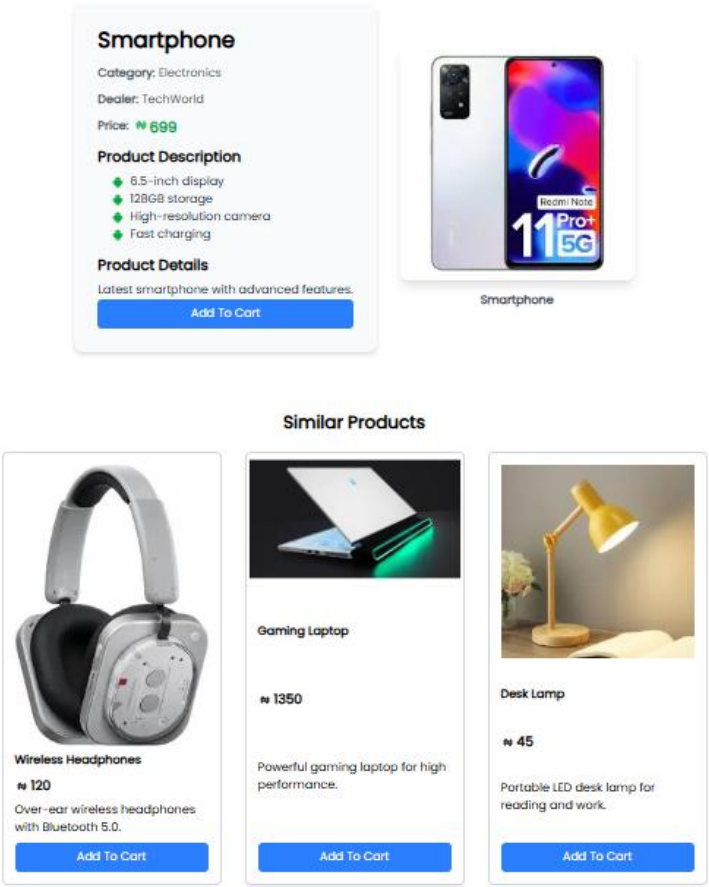


FIG 2.5: Product slug page & similar products

```
export default function ProductDetailsPage({ params }) {
      <div className="flex items-center justify-center gap-8 p-8">
        <section className="flex flex-col bg-gray-50 p-8 rounded-xl shadow-md">
          <h1 className="text-3xl font-bold mb-4">{productInfo.name}</h1>
          <p className="mb-3 text-gray-600">
            <span className="font-semibold">Category:</span>{" "}
            {productInfo.category}
          </p>
          <p className="mb-3 text-gray-600">
            <span className="font-semibold">Dealer:</span> {productInfo.dealer}
          </p>
          <p className="flex mb-3 text-gray-600 gap-2">
            <span className="font-semibold">Price: </span>{" "}
            <span className=" flex text-xl font-bold text-green-600">
              <TbCurrencyNaira />
              {productInfo.amount}
            </span>
          </p>
          <h2 className="text-xl font-semibold mb-2">Product Description</h2>
          <ul className="mb-4 pl-5 text-gray-800">
            {Array.isArray(productInfo.productDescription) ? (
              productInfo.productDescription.map((desc, idx) => (
                <li key={idx} className="flex items-center gap-2">
                  <DiAndroid className="text-green-600" /> {desc}
                </li>
              ))
            ) : (
              <li className="flex items-center gap-2">
                <DiAndroid className="text-green-600" />{" "}
                {productInfo.productDescription}
              </li>
            )}
          </ul>
```

2.1.7 Capstone

Capstone projects: At the end of the frontend section of the program, students worked on a Capstone project using React, Javascript, Tailwind CSS, displayed all the projects in a mini portfolio built with Next.js , deployed them, and presented in a demo session.

hosted link : https://mini-projects-portfolio-chi.vercel.app/

GitHub repository links: https://github.com/Victor-Olumide/mini-projects-portfolio

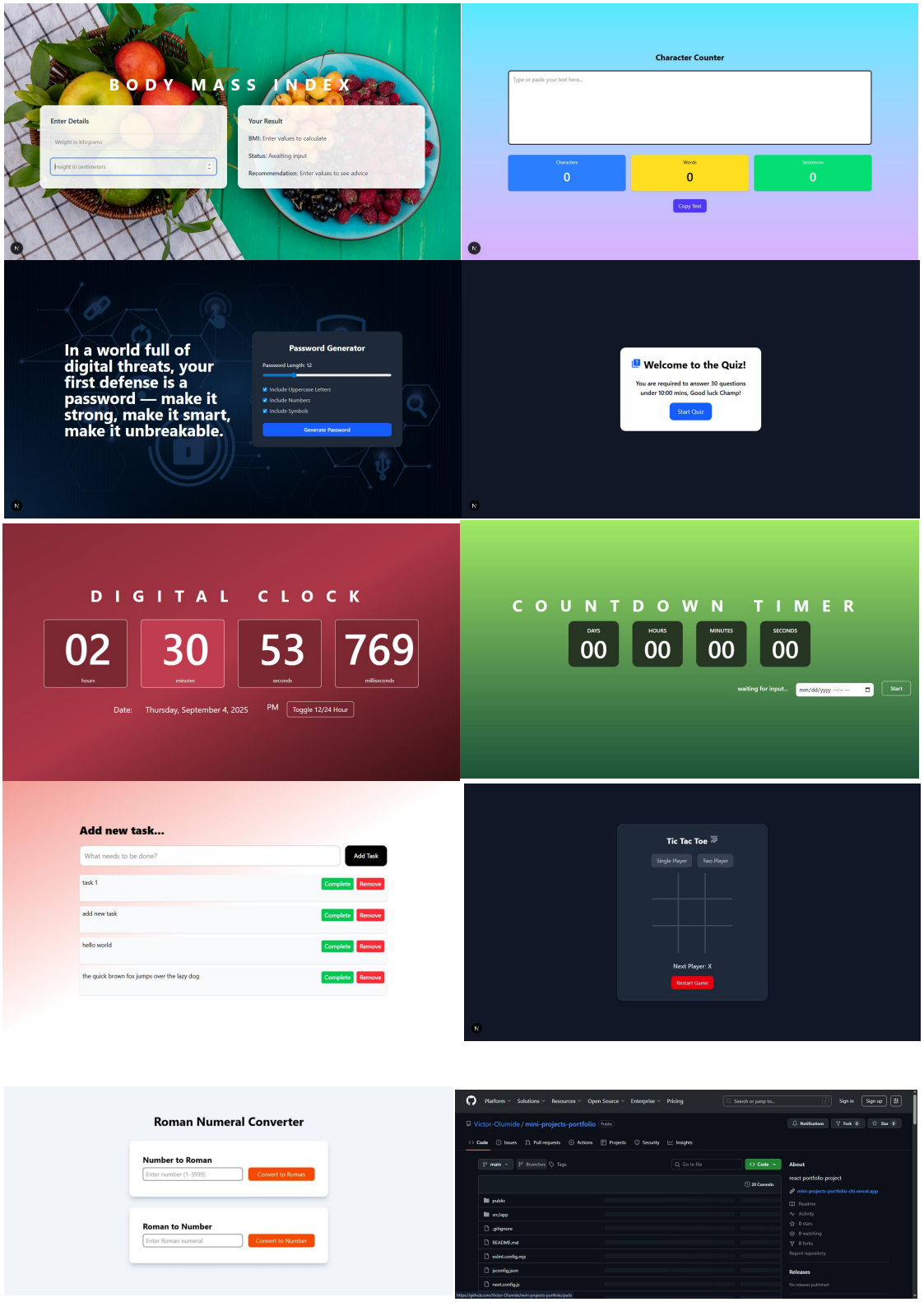


FIG 2.5: Capstone project screenshots & GitHub repo page

2.2 BACKEND DEVELOPMENT

2.2.1 Backend Fundamentals

Concepts taught: role of backend (business logic, data persistence), APIs (RESTful principles), servers, middleware, and HTTP status codes classification (2xx success, 4xx client error, 5xx server error).

2.2.2 Express & Server Setup

Packages installed: express, nodemon (dev utility), mongoose (ODM).

Typical install commands:

- npm init -y
- npm install express
- npm install -g mongoose
- npm install dotenv
- npm install nodemon
- npm install eslint

Core patterns learned: app setup, routing, req and res objects, route handlers, and status codes.

Dynamic routing: app.get('/products/:id', handler) and route parameter handling (req.params.id).

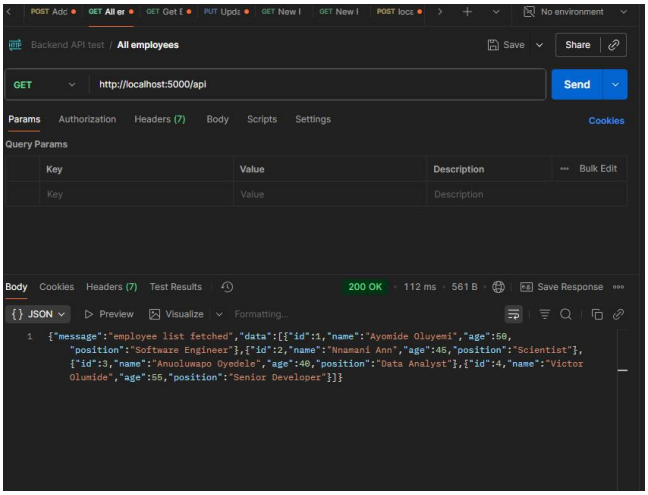


FIG 2.6: Postman GET request screenshot

2.2.3 Testing HTTP & CRUD

Tools used: Postman and HTTPie for API testing.

Tasks: tested POST, GET, PUT, DELETE endpoints; validated request/response bodies; implemented CRUD (Create, Read, Update, Delete) in routes and controllers.

Example: create user (POST /api/auth/sign-up), login (POST /api/auth/log-in), create subscription (POST /api/subscriptions), fetch subscriptions (GET /api/subscriptions).

2.2.4 Project: Subscription Tracker

Project goals: build a backend app to manage users and subscriptions with authentication, authorization, and subscription lifecycle.

Tech stack: Node.js, Express, MongoDB (Atlas), Mongoose, JWT for authentication, dotenv for environment variables, HTTPie/Postman for testing.

Cluster & DB: created a MongoDB Atlas cluster and database for the project.

Auth flow: registration → hash password (bcrypt) → sign token (JWT) → protect routes using verifyToken middleware.

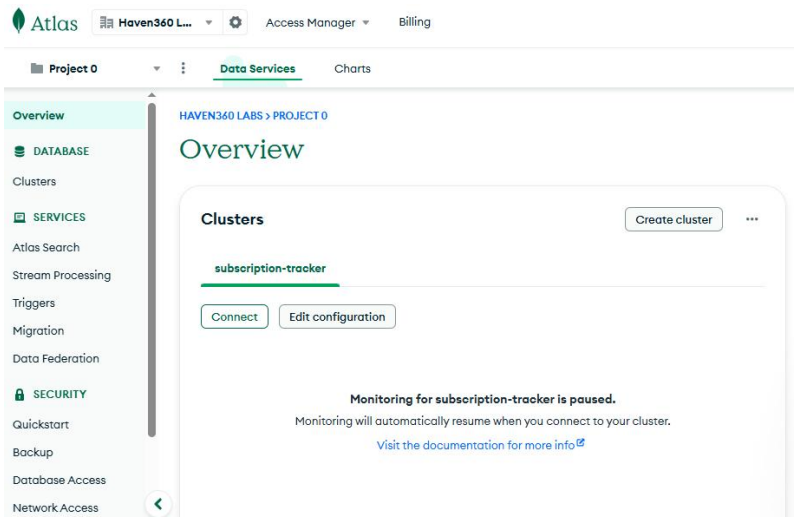


FIG 2.7: MongoDB Atlas cluster dashboard

2.2.5 Additional Tooling & Security

Installed tools: HTTPie (CLI), Arcjet (project scaffolding/automation tool), ESLint for code linting.

Security: JWT for stateless authentication, environment variables via .env (never commit secrets), basic rate-limiting/best practices discussed.

2.3 Assignments, Demos & Review Days

Demo day (Day 45): each student presented their frontend capstone — live demonstration of hosted projects and code walkthrough. Corrections were noted and made by the instructor and everyone was graded accordingly

CHAPTER THREE

WORK DONE AND EXPERIENCE GAINED

3.0 Introduction

This chapter presents a detailed account of the work carried out during the SWEP 2 program at Univelcity, highlighting the technical tasks, assignments, group projects, and capstone activities. It also describes the skills and competencies gained in the process, with emphasis on both technical knowledge and professional development.

The work covered in this program was divided into two major phases:

1. **Frontend Development** — focusing on user interface (UI) technologies (HTML, CSS, JavaScript, React, Tailwind CSS).

2. **Backend Development** — focusing on server-side technologies (Node.js, Express, MongoDB, Mongoose, JWT authentication).

3.1 Frontend Development Experience

3.1.1 HTML & CSS Foundations

Tasks: Built semantic web pages using HTML5 tags (<header>, <section>, <article>, <footer>), explored multimedia elements (<audio>, <video>), and practiced form handling (<input>, <textarea>, <select>). CSS was applied for styling, including color models (RGB, HSL), the box model, and responsive typography using relative units (rem, %, vh, vw).

Projects/Assignments: Created practice layouts such as country flags using CSS colors, styled forms, and wireframes for a portfolio website.

Skills Gained: Structuring accessible HTML, applying CSS selectors (universal, class, ID, child, sibling), and using responsive design techniques.

```
You, last week | 1 author (You)
import Image from "next/image";        You, 3 weeks ago • feat: rename file …
import Link from "next/link";
import ProjectCard from "./components/ProjectCard";
import Header from "./components/Header";
import HomeCard from "./components/HomeCard";
import SocialLinks from "./components/SocialLinks";
import SkillsGrid from "./components/SkillsGrid";

export default function Home() {
  return (
    <>
      <Header
        src="/logo.png"
        message={"welcome to my mini projects portfolio"}
      />

      <div className="flex flex-col items-center justify-center w-full min-h-screen p-4 md:p-8">
        <HomeCard
          src="/me.jpg"
          title={"Junior Software Developer, Graphic Designer"}
          name={"Oyewumi Victor Olumide"}
          desc={
            " I am a passionate and dedicated junior software developer with a knack for creating efficient
          }
        />

        <section className="mt-16 w-full max-w-6xl">
          <h2 className="text-3xl font-bold text-center mb-8 text-gray-800">
            Featured Projects
          </h2>
          <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
            <ProjectCard
```

FIGURE 3.1: Screenshot of React portfolio skeleton.

3.1.2 Layout Systems & Tailwind CSS

Tasks: Learned CSS Grid (grid-template-columns, grid-area) and Flexbox for layout management. Transitioned to Tailwind CSS for utility-first styling. Installed and configured Tailwind (npx tailwindcss init -p), applied utility classes, and defined custom components using @layer base and @layer components.

Projects: Designed responsive layouts, including product cards and grid-based sections; built a homepage with a navigation bar and footer.

Skills Gained: Faster UI prototyping, reusable class-based styling, and responsive design consistency across breakpoints.

```
@import "tailwindcss";


@layer components{
    .navbar{
        @apply bg-black text-white p-4 flex flex-col md:flex-row justify-between items-center h-screen
    }

    .navbar a {
        @apply hover:text-slate-800 hover:bg-gray-300 px-4 py-2 hover:font-bold transition-colors dura
    }

    .header{
        @apply flex flex-col items-center justify-center text-center p-8 bg-gray-100 rounded-lg h-screen
    }

    .header img{
        @apply w-64 h-64 border-y-2 rounded-full;
    }

    .header h1{
        @apply text-4xl font-bold uppercase text-slate-950;
    }

    .header p{
        @apply text-lg text-gray-700 max-w-2xl mx-auto md:text-2xl;
    }
}
```

FIGURE 3.2: Screenshot of Tailwind-based component.

3.1.3 JavaScript & DOM Manipulation

Topics Covered:

- Variables and data types (primitive/non-primitive).
- String methods (slice, toUpperCase, replace, trim, etc.).
- Operators (arithmetic, comparison, logical, ternary).
- Arrays and methods (splice, map, filter, reduce).
- Functions (declaration, expression, arrow, constructor, anonymous).
- Control flow (if/else, switch), loops (for, while, do...while).
- DOM manipulation using querySelector, getElementById, innerText, innerHTML, and event handling.
- Error handling with try { } catch { }.

Assignments & Classwork:

- Built a Task Manager app (add/remove tasks, update DOM dynamically).
- Implemented a BMI (Body Mass Index) Calculator that validates inputs and outputs comments.
- Created a Countdown Timer and a Live Clock.
- Skills Gained: Writing clean ES6+ JavaScript, manipulating the DOM for interactivity, and implementing validation and error handling in web apps.

FIGURE 3.3: Screenshot of BMI Calculator and Task Manager app.

3.1.4 React Development

Tasks:

- Introduction to React and its component-based architecture.
- Setup using create-react-app.
- Implemented pages: Homepage and Product pages.
- Used React Router for navigation and slug routes (/product/:slug) to display product details with three random similar products.
- Built a cart feature with an "Add to Cart" button, styled with CSS Modules (table.module.css).
- Responsive navigation with a hamburger menu for mobile screens.
- Deployed apps on GitHub, Netlify, and Vercel.
- Group Work: Collaborated on a portfolio website using the Rivera template as a reference.
- Skills Gained: Building reusable React components, managing state, implementing dynamic routing, and deploying React apps.

3.1.5 Capstone & Demo Presentation

At the end of the frontend phase, all interns presented their capstone projects during a demo session. Projects showcased included personal portfolios, product listing pages, and functional UI applications. I successfully presented my project, which was hosted online and documented in GitHub.

Skills Gained: Presentation, project documentation, and ability to explain technical solutions to peers and mentors.

FIGURE 3.5: Screenshot of deployed frontend capstone project (Netlify/Vercel).

3.2 Backend Development Experience

3.2.1 Backend Basics & Express Setup

Topics Covered: Fundamentals of backend systems, APIs, servers, databases, middleware, and HTTP status codes.

Tasks: Installed Express, Nodemon, and Mongoose; constructed responses with req and res; tested endpoints in the browser; learned about HTTP request methods (GET, POST, PUT, DELETE).

Skills Gained: Setting up Express servers, handling requests/responses, and understanding HTTP workflows.

3.2.2 CRUD Operations & Postman Testing

Tasks: Installed and used Postman and HTTPie to test APIs. Implemented CRUD operations within Express routes, tested them using Postman collections.

Skills Gained: API testing and debugging, endpoint validation, and practical RESTful API design.

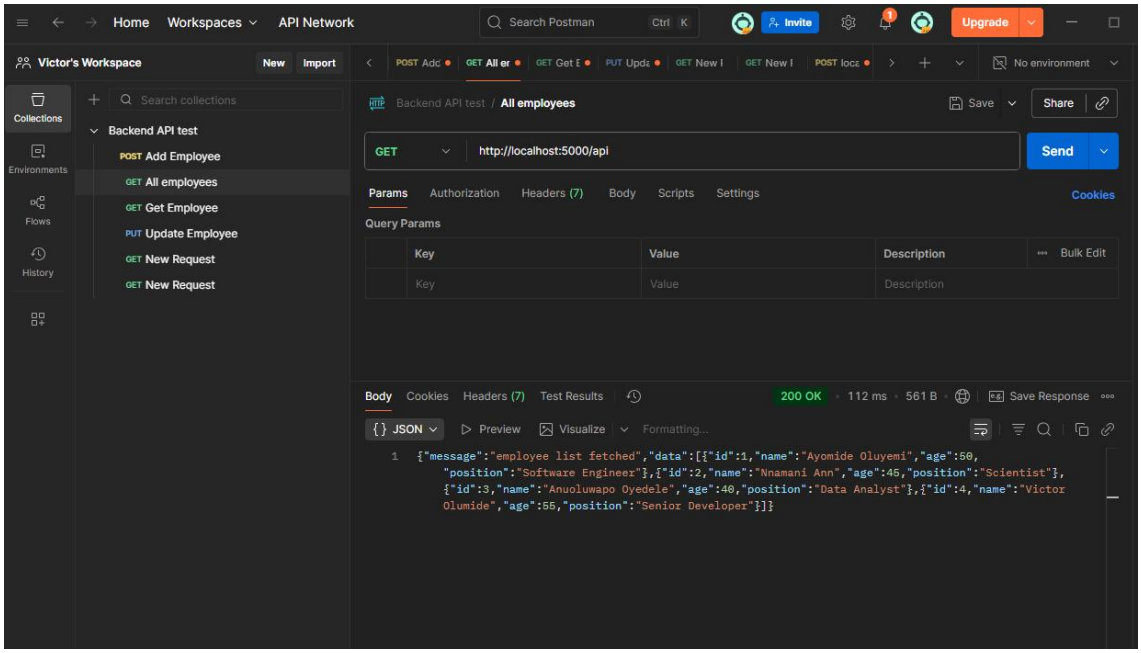


FIGURE 3.6: Postman screenshot of CRUD operations.

3.2.3 Subscription Tracker Project

Tasks: Built a Subscription Tracker backend project, structured as follows:

- Routes: auth.js, users.js, subscriptions.js.
- Models: User and Subscription models (Mongoose schemas).
- Controllers: Authentication, user, and subscription logic.
- Middleware: Error handler, authentication middleware using JWT.
- Database: MongoDB Atlas cluster created and connected.
- Technologies: Node.js, Express, MongoDB, Mongoose, JWT, dotenv, ESLint.

Skills Gained: Full backend project setup, secure authentication, database modeling, and error handling.

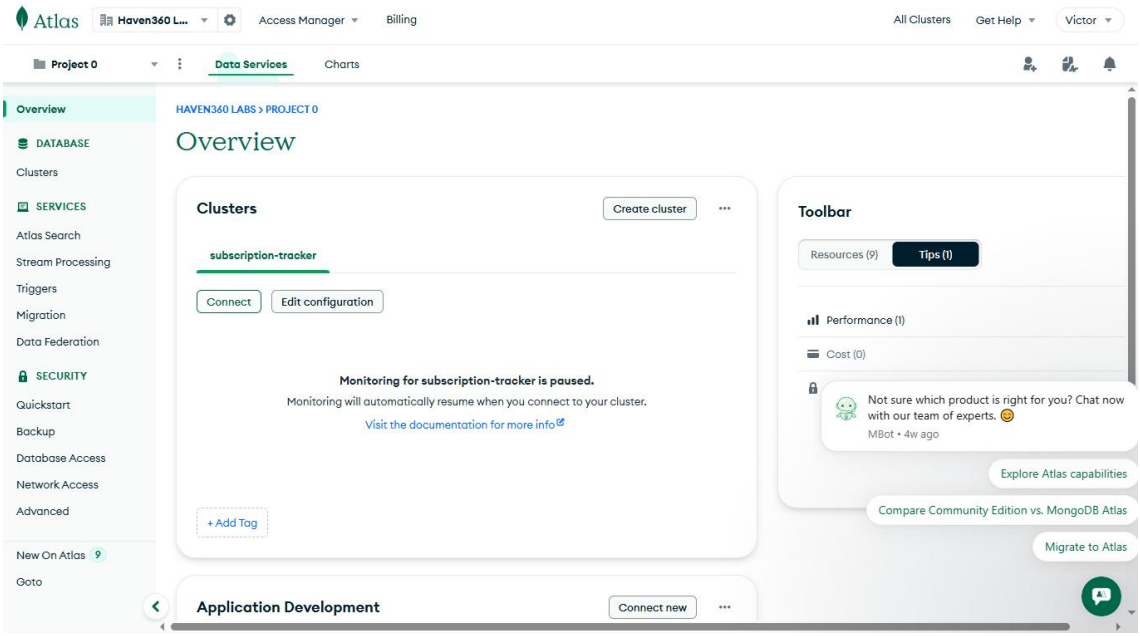


FIGURE 3.7: MongoDB Atlas dashboard screenshot of subscription cluster.

3.2.4 Integration & Frontend Connection

- Integrated the subscription tracker backend with frontend forms (React + Vite) to allow registration, login, and subscription management.
- Applied environment variables for secure configurations.
- Documented APIs and maintained repositories on GitHub.

Skills Gained: Connecting frontend and backend, environment security, and writing maintainable backend code.

3.3 Professional & Soft Skills Gained

Beyond technical expertise, the following professional skills were gained:

- Collaboration: Worked in groups, shared code on GitHub, and learned to manage team contributions.
- Problem-Solving: Debugged errors in servers, fixed dependency issues, and handled database connection problems.
- Time Management: Balanced daily tasks, assignments, and project deadlines.
- Presentation Skills: Presented projects during demo sessions and explained implementation clearly.
- Documentation: Maintained a structured logbook and documented endpoints in Postman.

3.4 Challenges and Solutions

- Challenge: Server errors during PostgreSQL setup earlier in the internship.

  Solution: Reinstalled PostgreSQL, configured ports, and reset authentication.

- Challenge: Handling eval() safely in JavaScript.

  Solution: Avoided unsafe use, relied on better parsing methods.

- Challenge: Authentication failures with JWT expiration.

  Solution: Implemented token refresh logic and proper error middleware.

CHAPTER FOUR

CONCLUSION, LIMITATIONS, AND RECOMMENDATIONS

4.0 Introduction

This chapter presents the conclusion of the SWEP 2 attachment, highlighting the overall experiences gained, lessons learned, challenges encountered, and recommendations for improving future training. The internship at Univelcity was an intensive, practice-oriented program that bridged academic knowledge with real-world software engineering skills.

4.1 Conclusion

The SWEP 2 attachment provided a rare opportunity to combine theoretical understanding with practical implementation in a structured, industry-oriented environment. Over the course of two months, the program transitioned from frontend development (HTML, CSS, JavaScript, React, Tailwind) to backend development (Node.js, Express, MongoDB, Mongoose, JWT authentication).

Each phase introduced progressively complex concepts, starting with simple static pages and culminating in fully functional applications such as the Subscription Tracker backend project and deployed React-based portfolio websites. Assignments, group projects, and demo sessions ensured that learning was continuous and performance-driven.

The experience has equipped me with both the technical competence to build and deploy web applications and the professional skills required to function effectively in collaborative, real-world environments.

4.2 Experience Gained

4.2.1 Technical Competence

Frontend Development:

- Built responsive user interfaces with semantic HTML5 and CSS3.
- Applied advanced layout techniques using Flexbox, CSS Grid, and Tailwind CSS.
- Developed interactive features with JavaScript, including DOM manipulation, form validation, and event handling.
- Created component-based user interfaces with React, implemented routing, and deployed applications using Netlify and Vercel.

Backend Development:

- Set up Express servers, implemented dynamic routing, and handled HTTP request/response cycles.
- Designed and tested RESTful APIs using Postman and HTTPie.
- Modeled and managed data using MongoDB Atlas and Mongoose.
- Implemented authentication and authorization using JWT, middleware for error handling, and secure environment configurations with dotenv.
- Built a Subscription Tracker project with routes, controllers, and database connectivity.

4.2.2 Professional Skills

Collaboration & Teamwork: Participated in group projects, shared responsibilities, and used Git/GitHub for version control and collaboration.

Problem-Solving: Learned to debug server issues, fix database connection errors, and resolve version conflicts in projects.

Time Management: Balanced daily lessons, assignments, and project submissions under tight deadlines.

Presentation Skills: Gained confidence through demo sessions, where I explained my projects and implementation strategies to peers and mentors.

Documentation: Maintained a daily logbook, API documentation in Postman, and well-structured GitHub repositories.

4.3 Challenges Encountered

Initial Setup Errors: Faced difficulties installing and configuring databases such as PostgreSQL and MongoDB, which delayed early backend work.

Authentication Issues: Token expiration and middleware bugs caused login failures in the subscription tracker project.

Time Constraints: The pace of daily lessons and assignments required careful prioritization to meet deadlines.

Limited Offline Practice: Some advanced backend concepts required stable internet connectivity for online resources, which posed occasional challenges.

4.4 Recommendations

4.4.1 For the Host Organization (Univelcity)

1. Provide more offline learning materials and resources so interns can continue practice without being dependent on internet connectivity.
2. Allocate additional review sessions at the end of each week to consolidate knowledge before moving on to new topics.
3. Incorporate more real-world projects that integrate both frontend and backend from the early weeks to build confidence in full-stack development.

4. Improve project demo sessions by inviting external reviewers (industry professionals) to give feedback and simulate real-world code reviews.

4.4.2 For Future Interns

1. Prepare adequately by reviewing HTML, CSS, and basic JavaScript before starting the program.
2. Develop good time management and documentation habits early, as daily tasks can accumulate quickly.
3. Embrace collaboration, since group projects provide a closer simulation of real-world development.
4. Actively ask questions and make use of mentors' feedback, as it accelerates learning.

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